

Roopansh Saxena

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Summary

Computer Science & Engineering student with hands-on experience in MERN stack development and AI-powered applications. National-level hackathon finalist, recognized for building innovative AI solutions to solve real-world engineering and workflow challenges. Experienced in developing full-stack applications, AI assistants, and intelligent automation systems. Passionate about software engineering, problem-solving, and continuously learning modern technologies to build scalable and impactful products.

Skills

Programming Languages: JavaScript (ES6+), C++ (Learning DSA)
Frontend: React.js, HTML5, CSS3, Tailwind CSS, Responsive Web Design
Backend: Node.js, Express.js, REST APIs, Authentication (JWT)
Database: SQL, MongoDB, Mongoose
AI & Machine Learning: LLM Application Development, Prompt Engineering, AI Agents, Retrieval-Augmented Generation (RAG), Reinforcement Learning Concepts, Reward Engineering, AI Workflow Automation
Tools & Platforms: Git, GitHub, Postman, VS Code, Hugging Face, Docker
Core CS Concepts: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Network
Soft Skills: Problem Solving, Analytical Thinking, Team Collaboration, Communication, Adaptability, Continuous Learning

Achievement

Finalist @ Meta X Pytorch Openenv Hackathon
AI RL LLM Hackathon
InboxWorld - Teaching LLM Survive Corporate Triage

- Built InboxWorld, an OpenEnv-compatible multi-agent environment that converts email triage from a static classification task into a long-horizon agent decision problem.
- Designed a four-agent decision pipeline with Inbox Analyst, Priority Planner, Responder Agent, and Supervisor Agent to handle prioritization, response generation, escalation, and safety checks.
- Integrated a Gradio demo UI with Hugging Face hosted LLM inference for natural email responses and a local multi-agent fallback for offline reliability.
- Built a minimal Hugging Face TRL training pipeline that converts environment transitions into prompt-action-reward records for fine-tuning small language models.

Technical Stack : Python, Gradio, Hugging Face Transformers, HF TRL, OpenEnv, Matplotlib, JSON, Multi-Agent Systems, Reward Modeling

Finalist @ Economic Times AI Hackathon 2.0
AI National Hackathon
EPC Guardian — AI Intelligence Platform for Data Centre EPC Project Delivery

- Built an offline-first EPC project intelligence platform for data centre delivery, focused on detecting specification deviations, predicting schedule impact, and supporting evidence-backed engineering decisions.
- Developed a full-stack prototype with 8 modules: project overview, document Q&A, compliance review, schedule impact, commissioning readiness, supply chain alerts, evidence library, and validation dashboard.
- Designed the system with offline fallback by default and optional API-key-based AI and RAG extension for future LLM-powered grounded answers and structured extraction.
- Validated the prototype with automated tests and a labelled retrieval benchmark covering compliance logic, search accuracy, commissioning blockers, schedule prediction, and supply-chain severity.

Technical Stack : JavaScript, Node.js, HTML/CSS, JSON, REST APIs, Rule-Based AI, RAG-style Retrieval, OpenAI API-ready architecture, Node.js Testing, Git/GitHub.

Project

MyGPT - MERN AI Assistant
Full-Stack AI Application

- Built a full-stack MERN AI assistant using React.js, Vite, Tailwind CSS, Express.js, MongoDB, and Mongoose.
- Implemented secure user authentication with JWT, bcrypt password hashing, protected routes, and persistent login state.
- Integrated a configurable AI provider layer supporting Demo mode, OpenAI API, and local Ollama for flexible development and deployment.

Technical Stack: React.js, Vite, Tailwind CSS, Node.js, Express.js, MongoDB, Mongoose, JWT, bcrypt, Axios, React Router, REST APIs, OpenAI API, Ollama, Git, GitHub

Education

Ajay Kumar Garg Engineering College B.Tech - Computer Science and Engineering 70%	Ghaziabad, Uttar Pradesh Oct 2023
Bishop Conrad Senior Secondary School Class 12 - PCM 91.4%	Bareilly, Uttar Pradesh Jun 2022
Bishop Conrad Senior Secondary School Class 10 88.8%	Bareilly, Uttar Pradesh Jun 2020